

SUMMARY TABLE: CHEMISTRY GRADE 10

Sub-Strand 1.5: Periodicity — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 1.5: Periodicity
Driving Question	DRIVING QUESTION: "Why do elements that sit next to each other in the periodic table — like sodium and magnesium, or chlorine and argon — behave so differently, and can we use an element's position in the table to predict exactly how it will behave before we ever put it in a test tube?"

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Something Strange at the Water Plant	Students observed video footage and demonstrations of Group I metals (lithium, sodium, potassium) reacting with water. Key observations included: Li reacts steadily; Na reacts vigorously with fizzing and movement across the water surface; K reacts explosively with a lilac flame. Students also recorded physical data (atomic radius, melting point, density) from data tables, noting clear numerical trends down the group. Pedagogical note: Cold-call at least 3 students to describe what 'trend' means before moving to explanation — many learners conflate 'trend' with 'pattern' without understanding directionality.	Students learned that Group I metals (alkali metals) show clear, predictable trends going DOWN the group: (1) atomic radius INCREASES as each successive element has an additional electron shell; (2) melting point DECREASES — Li melts at 181 °C, Na at 98 °C, K at 63 °C — because the metallic bond weakens as the atomic radius grows and nuclear attraction to delocalised electrons diminishes; (3) density generally INCREASES (Li 0.53 g/cm³ → K 0.86 g/cm³), though all three float on water; (4) reactivity with water INCREASES down the group because the outermost electron is further from the nucleus and more easily lost. These trends are not random — they are directly caused by increasing atomic number and electron shell count.	The trends observed in Group I are explained by electron configuration and nuclear shielding. Going down Group I, each element has one more complete electron shell, increasing atomic radius. The single outer (valence) electron is progressively further from the nucleus AND shielded from nuclear attraction by more inner electron shells — this is the shielding effect. As a result, less energy is needed to remove the outer electron (lower ionisation energy), making each successive metal more reactive. This directly answers a fragment of the driving question: elements in the SAME group behave SIMILARLY (all react with water) but with measurable differences in intensity because of subtle changes in electron arrangement. Teacher note: Draw an annotated diagram of Li vs K electron shells on the board and ask students to predict where Rb would fall before revealing data — this builds the habit of predictive reasoning central to the driving question.

Lesson 2 Sodium vs. Magnesium — Same Period, Different Worlds	Students observed side-by-side demonstrations or video clips of sodium and magnesium reacting with water and with dilute hydrochloric acid. Key observations: Na reacts explosively with cold water; Mg reacts only very slowly with cold water (a few bubbles over several minutes) but reacts vigorously with dilute HCl with rapid hydrogen gas evolution. Students also examined flame test results and recorded differences in physical appearance (Na is soft and silvery; Mg is harder and shinier). Pedagogical note: Prompt students to record observations using precise scientific language — 'vigorous effervescence' rather than 'it bubbled a lot' — before moving to explanation.	Students learned that sodium (Group I, Period 3) and magnesium (Group II, Period 3) sit next to each other in the same period yet behave very differently because they have different numbers of valence electrons: Na has 1 valence electron; Mg has 2. To achieve a stable noble-gas configuration, Na needs to lose only ONE electron, while Mg must lose TWO. Losing a second electron from Mg requires overcoming a higher second ionisation energy, making Mg less reactive than Na overall. Students also learned that within the same period, going LEFT to RIGHT, metallic character decreases and ionisation energy generally increases because the nuclear charge increases (more protons) while the number of shielding inner shells stays the same — so the nucleus holds outer electrons more tightly.	The contrasting behaviour of Na and Mg is explained by the concept of ionisation energy and the number of valence electrons. Sodium's single outer electron experiences relatively weak nuclear attraction (shielded by 2 inner shells) and is lost easily — low first ionisation energy. Magnesium has one more proton, pulling electrons inward more strongly, AND requires a second ionisation step to reach stability. This two-step energy cost makes Mg significantly less reactive with water, though still reactive enough to dissolve in acid. This lesson deepens the driving question answer: position in the PERIOD (not just the group) governs behaviour through changes in nuclear charge and ionisation energy. Teacher note: Use the analogy of a Nairobi matatu conductor collecting fares — Na only needs to collect one fare to be satisfied; Mg must collect two, and the second passenger is harder to convince.
Lesson 3 Magnesium, Calcium and the Stubborn Metals	Students observed reactions of Group II metals — magnesium (Mg), calcium (Ca) — with cold water and with dilute acid. Key observations: Mg reacts very slowly with cold water (barely visible bubbles) but rapidly with steam; Ca reacts steadily with cold water, producing Ca(OH)_2 (limewater turns milky when CO_2 is bubbled through it in a follow-up test). Students also examined the physical hardness of Group II metals compared to Group I, noting Group II metals are harder and have higher melting points. Pedagogical note: Allow at least 90 seconds of wait time after the Ca + water demo before asking for observations — students need time to notice the slower, steadier reaction compared to Na.	Students learned: (1) Group II metals are less reactive than Group I metals in the same period because they must lose TWO electrons rather than one, requiring higher combined ionisation energies; (2) going DOWN Group II (Mg $\rightarrow$ Ca $\rightarrow$ Sr $\rightarrow$ Ba), reactivity with water INCREASES for the same reason as in Group I — increasing atomic radius and shielding reduce the energy needed to remove both valence electrons; (3) Group II metals are harder, denser, and have higher melting points than their Group I neighbours because two delocalised electrons per atom create stronger metallic bonds; (4) the hydroxides of Group II metals (e.g., Ca(OH)_2) are only slightly soluble, unlike the fully soluble hydroxides of Group I metals.	Group II trends are explained by the same underlying principles as Group I — increasing atomic radius, increasing shielding, decreasing ionisation energy going down the group — but modified by the requirement to remove two valence electrons. The higher charge density of Group II ions (2+ compared to 1+) also means stronger ionic bonds in compounds and stronger metallic bonds in the pure metal, accounting for their greater hardness and higher melting points. This lesson refines the driving-question answer: it is not just GROUP membership but the NUMBER of valence electrons within a group that sets the baseline for reactivity and physical properties. Teacher note: Have students update their Driving Question Board (DQB) by adding a new sticky note: 'The number of valence electrons matters as much as the group number.' Connect to Lesson 2 observations about Na vs Mg.
Lesson 4	Students observed (via sealed sample	Learners learned that: (1) halogens exist in	Unlike metals (which react by LOSING

The Halogen Family: From Gas to Solid	tubes or video) the physical states and colours of the halogens at room temperature: Cl_2 is a pale yellow-green gas; Br_2 is a red-brown liquid; I_2 is a grey-black solid that sublimates to a purple vapour when gently heated. Students also observed displacement reactions: Cl_2 water displacing Br^- ions (orange colour appears in cyclohexane layer); Br_2 water displacing I^- ions (violet colour appears). Pedagogical note: The displacement reactions are a highlight — cold-call students to rank the halogens by reactivity based ONLY on what they see, before any explanation. Expect and welcome incorrect initial rankings as productive starting points.	all three physical states at room temperature — Cl_2 (gas), Br_2 (liquid), I_2 (solid) — with colour deepening down the group; (2) going DOWN Group VII, atomic radius increases, and the outer electron shell is further from the nucleus and more shielded, making it harder for the atom to attract an additional electron — therefore reactivity DECREASES down the group ($\text{Cl} > \text{Br} > \text{I}$); (3) a more reactive halogen can displace a less reactive halogen from its salt solution (Cl_2 displaces Br^-; Br_2 displaces I^-; Cl_2 displaces I^-); (4) boiling points and melting points INCREASE down the group because larger electron clouds create stronger van der Waals (London dispersion) forces between molecules.	electrons), halogens react by GAINING one electron to complete their outer shell. Reactivity therefore depends on how strongly the atom can attract an incoming electron — its electron affinity. Going down Group VII, the increasing atomic radius and shielding effect mean the outer shell is less effective at pulling in a new electron, so reactivity decreases. This is the OPPOSITE trend to Group I and II metals, which become MORE reactive going down. This contrast is a critical answer to the driving question: position in the periodic table predicts reactivity, but the DIRECTION of the trend depends on whether the element reacts by losing or gaining electrons. Teacher note: Explicitly draw a two-column comparison on the board: 'Metals going down → more reactive' vs 'Halogens going down → less reactive.' Ask students WHY the directions are opposite before explaining — this is a high-leverage conceptual moment.
Lesson 5 The Nobles and the Patterns of Eight	Students examined data tables showing the electron configurations of noble gases (He: 2; Ne: 2,8; Ar: 2,8,8; Kr: 2,8,8,18) and reviewed evidence for their chemical inertness — noble gases do not form compounds under normal conditions, do not react with water, acids, or metals, and were historically called 'inert gases.' Students may also have viewed a short video showing noble gas applications (neon signs glowing in Nairobi's CBD, argon used in welding, helium in weather balloons released from Wilson Airport). Pedagogical note: Begin with a provocative question — 'If chlorine (next door to argon) is so reactive, why does argon do absolutely nothing?' Give 3 minutes of think-pair-share before taking responses.	Group VIII (noble gas) elements are chemically inert because their outer electron shells are completely full — He has a full first shell (2 electrons); Ne, Ar, Kr have a full outer shell (8 electrons) — giving them maximum stability with no tendency to gain, lose, or share electrons. Students learned the 'octet rule': atoms of most elements react in ways that give them 8 electrons in their outer shell, mimicking the stable configuration of a noble gas. This explains WHY other elements react at all — they are striving toward the noble gas configuration. Students also learned that noble gases have very low boiling points (Ar boils at -186°C) because the only intermolecular forces are very weak van der Waals forces between non-polar atoms.	The inertness of noble gases is explained by their complete valence electron shells, which confer a uniquely low potential energy state. There is no energetic 'incentive' for a noble gas atom to react — gaining or losing electrons would disrupt a perfectly stable configuration. This is the conceptual anchor for the entire sub-strand: ALL the reactivity trends observed in Groups I, II, and VII can be understood as elements attempting to achieve the stable noble-gas electron configuration. Sitting right next to a noble gas in the periodic table — as Na sits next to Ne, or Cl sits next to Ar — does NOT mean they share properties; it means Na has ONE electron too many and Cl has ONE electron too few to reach that stable state, driving their very different and vigorous chemistries. This is a direct, full answer to the driving question. Teacher note: Return explicitly to the driving question text at this point. Ask students: 'Can you now explain why Na and Ar behave so differently even though they are

			in the same period?' Expect full-sentence answers referencing electron configuration.
Lesson 6 Crossing the Period — Trends Across Period 3	Students examined a data set for Period 3 elements (Na, Mg, Al, Si, P, S, Cl, Ar) covering: atomic radius, first ionisation energy, electronegativity, melting point, and electrical conductivity. They plotted graphs of first ionisation energy vs. atomic number across Period 3, observing a general INCREASE from Na to Ar (with two notable dips at Al and S). Students also observed or reviewed demonstrations of Period 3 oxides dissolving in water: $\text{Na}_2\text{O} \rightarrow$ strongly alkaline solution; $\text{MgO} \rightarrow$ weakly alkaline; $\text{Al}_2\text{O}_3 \rightarrow$ amphoteric; $\text{SiO}_2 \rightarrow$ insoluble; $\text{P}_4\text{O}_{10} \rightarrow$ acidic; $\text{SO}_3 \rightarrow$ strongly acidic; $\text{Cl}_2\text{O}_7 \rightarrow$ very strongly acidic. Pedagogical note: The graph-plotting activity is essential — do not skip it. Students who plot the data themselves retain the ionisation energy trend far better than those who only view a pre-drawn graph. Allow 10 minutes for individual plotting before class discussion.	Students learned that across Period 3 (left to right): (1) atomic radius DECREASES because nuclear charge increases (more protons) while the number of electron shells stays the same, pulling electrons closer; (2) first ionisation energy generally INCREASES because the stronger nuclear charge holds outer electrons more tightly; (3) metallic character DECREASES and non-metallic character INCREASES — Na and Mg are metals, Al is a metalloid, Si is a semiconductor, and P, S, Cl are non-metals; (4) the acid-base character of oxides shifts systematically from strongly basic (Na_2O) through amphoteric (Al_2O_3) to strongly acidic (Cl_2O_7), reflecting the change from metal to non-metal across the period; (5) electronegativity increases across the period for the same reason as ionisation energy.	The across-period trends are unified by a single cause: increasing nuclear charge (atomic number) with a constant number of electron shells. More protons pull the electron cloud inward (smaller radius), hold valence electrons more tightly (higher ionisation energy), and attract bonding electrons from other atoms more strongly (higher electronegativity). This converts elements progressively from electron-donors (metals) to electron-acceptors (non-metals) as one moves left to right. The oxide acidity trend is a chemical consequence of this shift — metal oxides are ionic and produce OH^- in water (basic); non-metal oxides are covalent and produce H^+ (acidic). This lesson completes the two-dimensional picture of the periodic table: GROUP trends (vertical) and PERIOD trends (horizontal) together allow prediction of any main-group element's behaviour. Teacher note: Emphasise that this lesson provides the final tool needed to fully answer the driving question — students can now predict behaviour from position in both dimensions of the table.
Lesson 7 What Are These Elements Good For? — Uses, Applications & Final Explanation	Students examined real-world and Kenyan-context applications of elements from Groups I, II, VII, and VIII: sodium compounds in food preservation (sodium chloride — table salt used in nyama choma seasoning, sodium carbonate — washing soda); calcium compounds in construction (calcium oxide/quicklime used in Nairobi building sites, calcium carbonate in cement); chlorine in water treatment (Nairobi City Water & Sewerage Company uses chlorination); iodine in antiseptics and iodised salt (addressing goitre in Rift Valley communities); argon in welding (Jua Kali artisans in Kamukunji); neon in commercial signage. Students also conducted a final structured review of all trend data collected across Lessons 1–6. Pedagogical note:	Students learned that an element's position in the periodic table — specifically its GROUP number (valence electrons, type of reactivity) and PERIOD number (atomic radius, ionisation energy, metallic/non-metallic character) — determines its electron configuration, which in turn determines ALL of its chemical and physical properties. They learned that this predictive power is the central achievement of the periodic table: given only an element's position, one can predict its likely reactions, the formulae of its compounds, the properties of its oxides, and its practical applications — before ever conducting an experiment. Students also consolidated the distinction between periodic trends (repeat across each period) and group trends	The driving question is fully answered: elements next to each other in the same period (e.g., Na and Mg, or Cl and Ar) behave differently because even a single additional proton and valence electron produces a fundamentally different electron configuration — different number of valence electrons, different ionisation energy, different tendency to gain or lose electrons. The periodic table encodes these differences in position. An element's group tells you HOW it reacts (by losing electrons $\rightarrow$ metals; by gaining electrons $\rightarrow$ non-metals; by not reacting $\rightarrow$ noble gases) and its period tells you HOW STRONGLY it reacts (via atomic radius and ionisation energy trends). Together, group and period position provide a complete predictive

	This lesson is the payoff lesson — insist that every explanation students give must reference electron configuration or ionisation energy. Refuse vague answers like 'because it is in Group I.' Push for mechanistic explanations.	(consistent down each group).	framework. Teacher note: Close the unit by returning to the driving question on the DQB and asking students to write a 5-sentence 'final explanation' in their notebooks — this serves as an informal summative assessment. Look for correct use of: valence electrons, ionisation energy, atomic radius, shielding effect, and noble gas configuration as the target stable state.
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